Predictive Maintenance: Anomaly Detection

Mathematical Documentation

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Overview

This document provides the mathematical foundation for each of the nine anomaly detection methods implemented in `src/anomaly_detection.py`. All methods operate by comparing a **test dataset** $\mathcal{A}$ (actual robot operation with hidden anomalies) against a **training dataset** $\mathcal{T}$ (10,000 rows of confirmed healthy operation). Baselines are computed from $\mathcal{T}$ prior to detection.

Notation

$x_i^{(c)}$	Value of sensor column c at row i in $\mathcal{A}$
μ_c, σ_c	Mean and standard deviation of column c in $\mathcal{T}$
$x_{\max}^{(c),\mathcal{T}}$	Maximum value of column c in $\mathcal{T}$
N	Total number of rows (10,000)
w	Rolling window width (rows)
$\hat{\rho}(\tau)$	Sample autocorrelation at lag τ

Sensor groups used throughout:

$$\mathcal{C} = \{\texttt{Current_J0}, \ldots, \texttt{Current_J5}\} \quad \text{(6 joint current sensors)}$$
$$\mathcal{T}_{\text{cols}} = \{\texttt{Temperature_T0}, \texttt{Temperature_J1}, \ldots, \texttt{Temperature_J5}\} \quad \text{(6 temperature sensors)}$$
$$\mathcal{S} = \{\texttt{Speed_J0}, \ldots, \texttt{Speed_J5}\} \quad \text{(6 speed sensors)}$$

[EASY] Method 1: Electrical Overload

Physical Motivation

Each joint motor draws current proportional to torque. A sudden, sustained elevation well above the normal operating envelope indicates a motor being forced against an obstruction, a short circuit, or a driver fault.

Threshold Definition

For each column $c \in \mathcal{C}$, the anomaly threshold is set at three standard deviations above the training mean (the upper 3σ bound):

$$\theta_c^{\mathrm{OL}} = \mu_c + 3\sigma_c \tag{1}$$

Run Detection

Define the binary indicator sequence:

$$b_i^{(c)} = \mathbf{1}\left[x_i^{(c)} > \theta_c^{\mathrm{OL}}\right], \quad i = 1, \ldots, N \tag{2}$$

A **run** is a maximal contiguous subsequence of consecutive rows where $b_i^{(c)} = 1$. Formally, a run starting at row s and ending at row e satisfies:

$$b_s^{(c)} = b_{s+1}^{(c)} = \cdots = b_e^{(c)} = 1, \quad b_{s-1}^{(c)} = 0 \text{ (or } s = 1), \quad b_{e+1}^{(c)} = 0 \text{ (or } e = N) \tag{3}$$

Detection Criterion

Let $\ell^* = \max_{\text{runs}}(e - s + 1)$ be the length of the longest run. An overload anomaly is reported when:

$$\ell^* \geq 10 \tag{4}$$

The reported row range is $[s^*, e^*]$ corresponding to this longest run. The peak value $\hat{x} = \max_{i \in [s^*, e^*]} x_i^{(c)}$ is also reported.

Justification

The 3σ threshold keeps the false positive rate below 0.3% under a Gaussian noise model. The minimum run length of 10 rows rejects isolated spikes (which appear in $\mathcal{T}$ as single-row exceedances) and requires a physically meaningful sustained overload event.

[EASY] Method 2: Overheating

Physical Motivation

Joint temperatures normally remain within a stable operating band. A rapid temperature exceedance indicates cooling failure, abnormal friction, or excessive duty cycle.

Threshold Definition

For each column $c \in \mathcal{T}_{\text{cols}}$, the threshold is set 2 °C above the observed training maximum:

$$\theta_c^{\text{HT}} = x_{\max}^{(c),\mathcal{T}} + 2 \tag{5}$$

The 2 °C buffer accounts for minor measurement variance without masking a genuine temperature exceedance.

Detection Criterion

The anomalous row set is:

$$\mathcal{H}_c = \left\{ i \mid x_i^{(c)} > \theta_c^{\text{HT}} \right\} \tag{6}$$

An overheating anomaly is reported for column c when $\mathcal{H}_c \neq \emptyset$. The reported range is $[\min \mathcal{H}_c,\ \max \mathcal{H}_c]$ and the peak temperature is $\max_{i \in \mathcal{H}_c} x_i^{(c)}$.

[EASY] Method 3: Repeated Safety Stops

Physical Motivation

The robot's safety system triggers a protective stop (`Robot_ProtectiveStop` = `True`) when it detects an unexpected collision or force limit breach. Occasional stops are normal; a cluster of stops in a short window indicates a persistent obstacle or path planning fault.

Baseline Rate

Let $F_i = \mathbf{1}[\texttt{Robot_ProtectiveStop}_i = \texttt{True}]$. The training baseline stop rate is:

$$\lambda_0 = \frac{1}{N} \sum_{i=1}^{N} F_i^{(\mathcal{T})} \tag{7}$$

In the training data, $\lambda_0 \approx 0.0014$ (approximately 14 stops per 10,000 rows).

Cluster Identification

Let $\mathcal{I} = \{i : F_i^{(\mathcal{A})} = 1\}$ be the set of stop rows in the actual data. A **cluster** is a maximal subset of $\mathcal{I}$ where consecutive elements are within 20 rows of each other:

$$i_{k+1} - i_k \leq 20 \quad \forall\, i_k, i_{k+1} \text{ consecutive in cluster} \tag{8}$$

Let $\mathcal{K}^*$ be the largest such cluster by cardinality.

Detection Criterion

A repeated-stop anomaly is reported when:

$$|\mathcal{K}^*| \geq 5 \quad \text{and} \quad \frac{|\mathcal{K}^*|}{\max(\mathcal{K}^*) - \min(\mathcal{K}^*) + 1} > 10\,\lambda_0 \tag{9}$$

Tightest Sub-Window Reporting

To report the most precise row range, the densest sub-window of 10 consecutive stops within $\mathcal{K}^*$ is found:

$$[s^*, e^*] = \arg \min_{\substack{[a,b] \subseteq \mathcal{K}^* \\ |[a,b]| = 10}} (b - a) \tag{10}$$

The local density within this sub-window is:

$$\rho^* = \frac{10}{e^* - s^* + 1} \tag{11}$$

[MEDIUM] Method 4: Wear and Tear

Physical Motivation

Mechanical wear in bearings, joints, or transmission components causes a gradual increase in the current required to maintain the same motion. This manifests as a sustained shift in the rolling mean current away from the healthy baseline.

Rolling Mean

For column $c \in \mathcal{C}$, the 20-row rolling mean at row i is:

$$\bar{x}_i^{(c)} = \frac{1}{20} \sum_{k=i-19}^{i} x_k^{(c)}, \quad i \geq 20 \tag{12}$$

Deviation from Baseline

The deviation of the rolling mean from the training mean is:

$$d_i^{(c)} = \left| \bar{x}_i^{(c)} - \mu_c \right| \tag{13}$$

The set of anomalous rows for column c is:

$$\mathcal{W}_c = \left\{ i \mid d_i^{(c)} > 2\sigma_c \right\} \tag{14}$$

Detection Criterion

An anomaly is reported for the column with the highest peak deviation, provided there is a sustained violation:

$$c^* = \arg\max_{c \in \mathcal{C}} \max_i d_i^{(c)} \quad \text{subject to} \quad |\mathcal{W}_{c^*}| \geq 50 \tag{15}$$

Reporting only the single worst sensor avoids double-counting co-occurring load shifts on mechanically coupled joints.

[MEDIUM] Method 5: Sensor Failure

Physical Motivation

A joint that reports non-zero speed while drawing near-zero current is physically impossible — moving a loaded arm requires electrical power. This contradiction indicates a faulty or miscalibrated speed encoder.

Inconsistency Condition

For joint j, a row i is flagged as physically inconsistent when:

$$\left|x_i^{(\texttt{Speed_J}j)}\right| > 1.0 \quad \text{and} \quad \left|x_i^{(\texttt{Current_J}j)}\right| < 0.05 \tag{16}$$

Let $\mathcal{P}_j$ be the set of all such rows for joint j.

Cluster Identification

Within $\mathcal{P}_j$, a **dense cluster** $\mathcal{D}_j$ is the largest subset of consecutive indices with inter-element gaps of at most 5 rows:

$$i_{k+1} - i_k \leq 5 \quad \forall\, i_k, i_{k+1} \text{ consecutive in } \mathcal{D}_j \tag{17}$$

Detection Criterion

Let $n_j^{(\mathcal{T})}$ be the number of inconsistent rows for joint j in the training data (baseline noise floor). A sensor failure is reported when:

$$|\mathcal{D}_j| \geq \max\Big(20,\ 5 \cdot n_j^{(\mathcal{T})}\Big) \tag{18}$$

The factor of 5 over the training baseline and minimum count of 20 ensures only genuine clustered failures are reported, not isolated noise events that also appear in healthy data.

[MEDIUM] Method 6: Electrical Noise

Physical Motivation

Damaged wiring insulation, loose connectors, or electromagnetic interference (EMI) cause rapid, erratic fluctuations in current readings. This appears as abnormally high short-term variance.

Rolling Standard Deviation

For column $c \in \mathcal{C}$, the 20-row rolling standard deviation is:

$$s_i^{(c)} = \sqrt{\frac{1}{19} \sum_{k=i-19}^{i} \left(x_k^{(c)} - \bar{x}_i^{(c)} \right)^2}, \quad i \geq 20 \tag{19}$$

Detection Criterion

The set of noisy rows for column c is:

$$\mathcal{N}_c = \left\{ i \mid s_i^{(c)} > 3\sigma_c \right\} \tag{20}$$

An anomaly is reported for the column with the highest peak rolling std, provided the event is sustained:

$$c^* = \arg\max_{c \in \mathcal{C}} \max_i s_i^{(c)} \quad \text{subject to} \quad |\mathcal{N}_{c^*}| \geq 20 \tag{21}$$

The $3\sigma_c$ threshold is calibrated so that the rolling std of healthy data (which itself fluctuates) does not exceed this bound, while genuine noise bursts produce rolling std values significantly above it.

[HARD] Method 7: Control Issues

Physical Motivation

A well-tuned PID temperature controller produces smooth, bounded temperature traces. An unstable or oscillating control loop introduces periodic patterns into the temperature time series — identifiable via autocorrelation analysis.

Autocorrelation Function

For a time series segment $\{y_t\}_{t=1}^{w}$ with mean $\bar{y}$, the sample autocorrelation at lag τ is:

$$\hat{\rho}(\tau) = \frac{\sum_{t=1}^{w-\tau}(y_t - \bar{y})(y_{t+\tau} - \bar{y})}{\sum_{t=1}^{w}(y_t - \bar{y})^2} \tag{22}$$

Window Scan

For each column $c \in \mathcal{T}_{\text{cols}}$ and each starting row i (step size 25), the detrended actual and training segments of length $w = 75$ are:

$$\tilde{x}_{i:i+w}^{(\mathcal{A})} = x_{i:i+w}^{(c)} - \bar{x}_{i:i+w}^{(c)} - \hat{\beta}(i - \bar{i}) \tag{23}$$

$$\tilde{x}_{i:i+w}^{(\mathcal{T})} = x_{i:i+w}^{(c),(\mathcal{T})} - \bar{x}_{i:i+w}^{(c),(\mathcal{T})} - \hat{\beta}^{(\mathcal{T})}(i - \bar{i}) \tag{24}$$

where $\hat{\beta}$ is the OLS slope of a linear trend fit to the segment.

The maximum autocorrelation for each segment is:

$$A^{(\mathcal{A})}(i,c) = \max_{\tau \in \{3,4,\ldots,11\}} \hat{\rho}^{(\mathcal{A})}(\tau) \qquad A^{(\mathcal{T})}(i,c) = \max_{\tau \in \{3,4,\ldots,11\}} \hat{\rho}^{(\mathcal{T})}(\tau) \tag{25}$$

Detection Criterion

The ACF excess for window (i, c) is:

$$E(i,c) = A^{(\mathcal{A})}(i,c) - A^{(\mathcal{T})}(i,c) \tag{26}$$

An anomaly is reported at the window and sensor with the highest excess:

$$(i^*, c^*) = \arg\max_{(i,c)} E(i,c) \quad \text{subject to} \quad A^{(\mathcal{A})}(i^*, c^*) > 0.7 \;\text{ and }\; E(i^*, c^*) > 0.3 \tag{27}$$

Comparing against the training ACF in the same window controls for any intrinsic periodic structure in the robot's motion cycle, isolating genuine control loop instability.

[HARD] Method 8: Coordination Problems

Physical Motivation

Adjacent joints on a collaborative robot arm operate under a shared motion plan. When the mechanical coupling or control signals between two joints degrade, the joints begin working against each other: one moves forward while the other moves backward during the same motion event.

Co-Movement Definition

For adjacent joint pair (j_1, j_2), row i is a **co-movement event** when both joints are simultaneously active:

$$\mathcal{M}_{j_1 j_2} = \left\{ i \;\middle|\; \left| x_i^{(\texttt{Speed_J}j_1)} \right| > 0.5 \;\text{ and }\; \left| x_i^{(\texttt{Speed_J}j_2)} \right| > 0.5 \right\} \tag{28}$$

Baseline Opposite-Direction Rate

During co-movement events in training, the fraction moving in opposite directions (i.e. speed signals have opposite sign) defines the baseline:

$$\pi_0^{(j_1 j_2)} = \frac{1}{|\mathcal{M}_{j_1 j_2}^{(\mathcal{T})}|} \sum_{i \in \mathcal{M}_{j_1 j_2}^{(\mathcal{T})}} \mathbf{1}\left[x_i^{(\texttt{Speed_J}j_1)} \cdot x_i^{(\texttt{Speed_J}j_2)} < 0 \right] \tag{29}$$

For this robot, $\pi_0^{(1,2)} \approx 0.45$.

Rolling Window Scan

For each 300-row window starting at row i, the observed opposite-direction rate is:

$$\hat{\pi}(i, j_1 j_2) = \frac{\displaystyle\sum_{k \in \mathcal{M}_{j_1 j_2} \cap [i,\, i+300)} \mathbf{1}\left[x_k^{(\texttt{Speed_J}j_1)} \cdot x_k^{(\texttt{Speed_J}j_2)} < 0 \right]}{|\mathcal{M}_{j_1 j_2} \cap [i,\, i+300)|} \tag{30}$$

This is only computed when $|\mathcal{M}_{j_1 j_2} \cap [i, i+300)| \geq 5$.

Detection Criterion

The coordination excess for window $(i, j_1 j_2)$ is:

$$\Delta\pi(i, j_1 j_2) = \hat{\pi}(i, j_1 j_2) - \pi_0^{(j_1 j_2)} \tag{31}$$

An anomaly is reported at:

$$(i^*, j_1^*, j_2^*) = \arg\max_{(i,\, j_1 j_2)} \Delta\pi(i, j_1 j_2) \quad \text{subject to} \quad \Delta\pi > 0.3 \tag{32}$$

The reported row range $[r_{\min}, r_{\max}]$ is the span of the exact opposite-direction event rows within the best window:

$$r_{\min} = \min \left\{ k \in [i^*, i^* + 300) \cap \mathcal{M}_{j_1^* j_2^*} : x_k^{(\texttt{Speed_J}j_1^*)} \cdot x_k^{(\texttt{Speed_J}j_2^*)} < 0 \right\} \tag{33}$$

$$r_{\max} = \max \left\{ k \in [i^*, i^* + 300) \cap \mathcal{M}_{j_1^* j_2^*} : x_k^{(\texttt{Speed_J}j_1^*)} \cdot x_k^{(\texttt{Speed_J}j_2^*)} < 0 \right\} \tag{34}$$

[HARD] Method 9: Early Degradation

Physical Motivation

Early mechanical degradation (bearing fatigue, lubrication loss, micro-fractures) manifests as subtle increases in the RMS current drawn by a joint — higher than a healthy robot performing the same motion. This is detectable by comparing per-segment RMS values against the training baseline.

Root Mean Square Current

For column $c \in \mathcal{C}$ and a segment of $w = 15$ rows starting at row s:

$$\mathrm{RMS}_s^{(c)} = \sqrt{\frac{1}{w} \sum_{i=s}^{s+w-1} \left(x_i^{(c)}\right)^2} \tag{35}$$

The RMS captures both the mean load and the variance of the current signal, making it more sensitive to subtle load changes than the mean alone.

Per-Segment Comparison

The dataset is divided into non-overlapping 15-row chunks. For each chunk k and column c, the RMS difference against the same chunk in training is:

$$\delta_k^{(c)} = \left| \mathrm{RMS}_{kw}^{(c),(\mathcal{A})} - \mathrm{RMS}_{kw}^{(c),(\mathcal{T})} \right| \tag{36}$$

Detection Criterion

The anomaly is reported at the single chunk and sensor with the highest RMS deviation above the training baseline:

$$(k^*, c^*) = \arg\max_{(k,\,c)} \delta_k^{(c)} \quad \text{subject to} \quad \delta_{k^*}^{(c^*)} > 0.5\,\mathrm{A} \tag{37}$$

The reported row range is $[k^* w,\ (k^*+1)w - 1]$, giving a 15-row window. The 15-row granularity provides the tightest localisation possible while maintaining a statistically stable RMS estimate.

Detection Parameter Summary

#	Anomaly Type	Severity	Window	Key Threshold
1	Electrical Overload	EASY	Run-based	$\mu_c + 3\sigma_c$, run ≥ 10 rows
2	Overheating	EASY	Exact rows	$x^{\mathcal{T}}_{\max} + 2\,°\mathrm{C}$
3	Repeated Safety Stops	EASY	Cluster	gap ≤ 20, ≥ 5 stops, $> 10\lambda_0$
4	Wear and Tear	MEDIUM	20-row	$\lvert\bar{x}_i - \mu_c\rvert > 2\sigma_c$, ≥ 50 rows
5	Sensor Failure	MEDIUM	Cluster	gap ≤ 5, $\geq \max(20, 5n^{\mathcal{T}}_j)$ rows
6	Electrical Noise	MEDIUM	20-row	$s^{(c)}_i > 3\sigma_c$, ≥ 20 rows
7	Control Issues	HARD	75-row	ACF excess > 0.3, $A^{\mathcal{A}} > 0.7$
8	Coordination Problems	HARD	300-row	$\Delta\pi > 0.3$ opposite-direction excess
9	Early Degradation	HARD	15-row	$\delta^{(c)}_k > 0.5\,\mathrm{A}$ RMS vs training

Data-Driven Results on Actual Dataset

Running the detector against `actual_data.csv` (with `training_data.csv` as baseline) produces exactly 9 findings:

Type	Sensor	Row Range	Span
Electrical Overload	`Current_J1`	498 – 619	122 rows
Overheating	`Temperature_J3`	1340 – 1359	20 rows
Repeated Safety Stops	`Robot_ProtectiveStop`	2565 – 2601	36 rows
Wear and Tear	`Current_J1`	502 – 633	132 rows
Sensor Failure	`Speed_J4`	5199 – 5379	181 rows
Electrical Noise	`Current_J1`	502 – 635	134 rows
Control Issues	`Temperature_J1`	525 – 600	75 rows
Coordination Problems	`Speed_J1/J2`	8109 – 8390	282 rows
Early Degradation	`Current_J1`	510 – 524	15 rows